Dear Mr Isaac,

CONCEPT NOTE AND SCOPE GUIDANCE FOR THE APPLICATION PREPRATION FOR THE CROSS RIVER BLUE ID & FISHERIES INFORMATION MANAGEMENT SYSTEM-FISHERMEN DATABASE, BLUE ID REGISTRATION PORTAL AND OFFLINE DATA COLLECTION SYSTEM

I hope this meets you well.

Sequel to our conversation about the need to have an organized database of the maritime community in Cross River State for the intended roll out intervention program by the department of Blue Economy, please see below a concept note and scope guidance for your design and development a robust yet friendly digital platform for data collection and government engagement in the fishing community.

1. STRATEGIC RATIONALE

Cross River State has numerous fishing communities spread across riverine, coastal, estuarine and inland areas from Calabar, Odukpani, Akpabuyo to Bakassi. A significant number of fishermen and other actors within the fisheries value chain operate informally, making it difficult for government and development partners to know the exact number of active fishermen, their locations, fishing activities, equipment, production capacity, economic needs and access to government programmes. In fact, there is an absence of organized and reliable fisherman and community data.

There is therefore a need to develop a centralized digital database that will capture, organize and maintain reliable information on fishermen operating in all fishing communities across Cross River State. The proposed system shall assign every registered fisherman a unique identification number known as a **Blue ID**.

The Blue ID will become the primary identification reference for fishermen participating in government fisheries programmes, grants, equipment distribution, training, insurance, cooperatives, credit programmes and other blue-economy interventions.

2. OBJECTIVE OF THE PROJECT

The objective is to design and develop a secure, user-friendly fisheries information management platform capable of registering and managing fishermen across Cross River State. The platform should work in areas with reliable internet connectivity as well as remote fishing communities where internet access is poor or unavailable.

The system should therefore operate in both online and offline modes. A particularly important design decision is to make it offline-first rather than merely "offline capable." That will make the system practical in remote riverine communities where the registration team may have no network for an entire day.

3. PROPOSED SYSTEM SCOPE

The proposed system should consist of three major components:

A. Mobile Data Collection Application

An Android-based application to be used by authorized field officers, enumerators and fisheries officials to register fishermen in fishing communities.

The application must work completely offline and store information locally on the device until internet connectivity becomes available.

Once the device connects to the internet, all verified records should automatically synchronize with the central database.

B. Central Web-Based Administration Portal

A secure web portal should allow authorized government officials and administrators to view, manage and analyse the statewide fishermen database.

The portal should provide access according to user roles, such as: State Administrator, Ministry/Fisheries Department, Local Government Administrator, Field Enumerator, Cooperative Officer.

C. Blue ID Identification System

Every verified fisherman should automatically receive a unique permanent identification number referred to as a **Blue ID**.

An example format may be: **CRS-BLU-BAK-0001.** Where:

CRS = Cross River State
BLU = Blue Economy/Fisheries Identification
BAK = Local Government or community code
0001 = Unique fisherman registration number

4. EXPECTED OUTPUT

The application design should carry:

a) **Fisherman Registration Information:** where the system should capture information including, but not limited to:

- Blue ID
- Full name
- Passport photograph
- Gender
- Date or estimated year of birth
- Phone number
- National Identification Number, NIN (where available)
- Landing site
- Local Government Area
- Next of kin/Next-of-kin phone number

b) **Fisheries Information:** The database should capture the fisherman's operational information, including:

- Type of fisherman
- Artisanal fisherman
- Commercial fisherman
- Fish farmer
- Boat owner
- Crew member
- Fish processor
- Fish trader
- Fishing community
- Primary landing site
- Type of fishing undertaken
- Marine fishing
- River fishing
- Creek fishing
- Estuary fishing
- Inland fishing
- Aquaculture
- Number of years fishing
- Cooperative membership

c) **Boat and Fishing Equipment Database:** Where applicable, each fisherman should be able to register assets connected to his or her operations. These may include:

- Boat ownership
- Boat registration number
- Boat type
- Canoe type
- Boat length
- Boat photograph
- Outboard engine
- Engine manufacturer
- Engine horsepower
- Engine serial number
- Fishing nets
- Fishing gear
- Cold boxes
- Ice boxes
- Safety equipment
- GPS equipment

The system should allow more than one boat or engine to be connected to a single Blue ID.

d) **Offline Operation:** Offline capability is one of the most important requirements of this project. Many fishing communities may experience poor or nonexistent internet connectivity. The field application must therefore allow an enumerator to:

- Log into the application before traveling to the community.
- Register fishermen without internet access.
- Take and save photographs offline.
- Record fishing and boat information.
- Generate a temporary registration reference.
- Store records securely on the device.
- Synchronize the records automatically when internet becomes available.

No information should be lost because the device temporarily loses connectivity. The system should also prevent duplicate records when synchronization occurs.

The system must not depend on fishermen owning smartphones.

Registration should primarily be conducted through authorized government enumerators or designated community registration officers using any device obtainable.

After registration, fishermen may receive a physical Blue ID card containing:

- Fisherman's photograph
- Full name
- Blue ID number
- Fishing community
- Local Government Area
- QR code
- Issue date

The QR code should allow an authorized official to retrieve or verify the fisherman's record.

For fishermen using basic mobile phones, the system should also be designed to support future **SMS or USSD services**. For example, a fisherman could eventually dial a short code or send an SMS to:

- Confirm Blue ID status
- Receive government announcements
- Receive training information
- Receive programme notifications
- Confirm subsidy or equipment allocations

5. BLUE ID VERIFICATION

Each Blue ID should have a corresponding digital profile. When the ID number or QR code is entered or scanned, authorized users should be able to verify:

- Fisherman's identity
- Photograph
- Community
- Registration status
- Boat information
- Cooperative
- Programme eligibility
- Benefits received

The system should provide a visible status such as:

- Active
- Pending Verification
- Suspended
- Deceased
- Inactive
- Duplicate Record Under Review

6. GOVERNMENT PROGRAMME MANAGEMENT

A major purpose of the database will be to improve the targeting and monitoring of government interventions. The system should permit programmes to be created within the portal. Examples include:

- Outboard engine distribution
- Fishing-net distribution
- Boat distribution
- Life-jacket programmes
- Fuel subsidy programmes
- Grants
- Loans
- Fish-smoking equipment
- Cold-storage support
- Aquaculture inputs
- Insurance
- Training programmes

Every benefit issued should be connected to the fisherman's Blue ID. This will allow government to see exactly who received assistance, when it was received and under which programme. It should significantly reduce duplication and multiple registrations for the same government intervention.

7. DASHBOARD AND REPORTING

The administration portal should include an executive dashboard showing information such as:

- Total registered fishermen
- Fishermen by community
- Male and female fishermen
- Boat owners
- Number of registered boats
- Number of outboard engines
- Fish farmers
- Fish processors
- Fishermen by fishing type
- Cooperative membership
- Government interventions received
- Communities lacking cold storage
- Communities lacking landing facilities

- Communities lacking boat repair services

The system can display relevant information through tables, maps, graphs and charts. Reports should be exportable in formats such as:

- PDF
- Excel
- CSV

8. GIS AND MAPPING

Where GPS data are collected, the portal should eventually provide an interactive map showing fishing communities throughout Cross River State. Clicking a community on the map should display relevant statistics including:

- Number of fishermen
- Boats
- Engines
- Fish production
- Infrastructure
- Landing sites
- Cold storage
- Processing facilities
- Major development needs

This will support infrastructure planning and investment decisions.

9. DATA SECURITY

Because the system will contain personal information, security should be incorporated from the beginning a such, these elements should be included:

- Secure login
- Role-based permissions
- Encrypted data transmission
- Secure database hosting
- Regular automated backups
- User activity logs
- Audit trails
- Password controls
- Device authentication
- Data synchronization controls
- Duplicate-registration detection

10. DUPLICATE PREVENTION

The system should be capable of detecting potential duplicate registrations using a combination of Name, Phone number, Photograph, National Identification Number(where available), Date of birth, Community, Boat information.

Potential duplicate records should be flagged for administrative review rather than automatically deleted.

11. FUTURE EXPANSION OF MODULE

The platform should be developed with a modular structure so additional blue-economy programmes can eventually be added. Future modules may include:

- Fish landing statistics
- Daily fish prices
- Catch reporting
- Aquaculture farms
- Fish processors
- Fish traders
- Cooperatives
- Fish markets
- Cold-chain facilities
- Boat registration
- Outboard-engine registration
- Engine maintenance records
- Marine safety
- Weather alerts
- Fisheries licensing
- Grants and loans
- Insurance
- Fishing permits

The Blue ID would eventually become the common identification platform connecting these services.

12. PROPOSED MINIMUM TECHNOLOGY REQUIREMENT

Kindly propose an appropriate technical architecture. However, the system should preferably include:

- **Field Application:**
  Android mobile application with offline-first capability.
- **Web Portal:**
  Responsive web application capable of operating on computers, tablets and smartphones.
- **Database:**
  Secure cloud-based centralized relational database.
- **Synchronization:**
  Automatic two-way synchronization between field devices and the central server.
- **Hosting:**
  Cloud infrastructure with automated backup.
- **API:**
  A secure API architecture allowing future government systems and applications to connect to the Blue ID database.

13. MINIMUM VIABLE PRODUCT

The first version should preferably contain:

- User login and permissions.
- Fisherman registration.
- Automatic Blue ID generation.
- Photograph capture.
- Offline data collection.
- Data synchronization.
- Boat registration.
- Outboard-engine registration.
- Cooperative registration.
- Government intervention records.
- Search and Blue ID verification.
- Dashboard.
- Basic GIS mapping.
- Excel/PDF reporting.
- Audit trail.
- Database backup.

14. SUMMARY OF EXPECTED OUTCOME

The project should ultimately create a single authoritative digital registry of fishermen and fishing communities throughout Cross River State. For the first time, Blue Economy department should be able to answer questions such as:

- How many fishermen are operating in Cross River State?
- Where are they located?
- How many fishing boats exist?
- How many outboard engines are operating?
- Which communities need landing sites?
- Which communities need cold rooms or ice plants?
- Who has received government assistance?
- Which fishermen qualify for a particular intervention?
- How much fishing capacity exists in each LGA?

This information will allow government to move from generalized fisheries interventions toward data-driven blue-economy planning and targeted investment.

This concept note is intended to provide clarity of expectation while allowing you as the developer deploy your professional discretion to design a system architecture that reflects the overall vision of collecting, and managing relevant data in the fishing community by the department of Blue Economy and relevant stakeholders going forward in Cross River State.

We look forward to your proposed technical solution covering all areas of raised and assure you of our gratitude and availability for any further engagement to deliver this landmark Blue Economy project under the leadership of the SA Blue Economy.

Please note the application should be simple enough for field officers to use in remote fishing communities while being sufficiently robust to become the long-term Cross River State Fisheries and Blue Economy Information Management System.

Thank you.

Yours sincerely,



Engr Diana Nsemo
Consultant to SA Blue Economy